Question 1:

Find the solution of the second-order differential equation $16x^2y'' - 8xy' + 9y = 0$ for $0 < x < \infty$.

a) The singular point of the differential equation $16x^2 \cdot y'' - 8x \cdot y' + 9y = 0$ is ___?___

b) The characteristic (or auxiliary) equation is ___?___

Hint: Let $y = x^r$. Then $y' = rx^{r-1}$ and $y'' = (r-1)rx^{r-2}$.

Substitute y, y' and y'' into $16x^2y'' - 8xy' + 9y = 0$: $16x^2\left[(r-1)rx^{r-2}\right] - 8x\left[rx^{r-1}\right] + 9x^r = 0$; factor out the common term x^r.

c) The roots of characteristic (or auxiliary) equation are ___?___

d) The general solution of the differential equation $16x^2 \cdot y'' - 8x \cdot y' + 9y = 0$ is ___?___

Question 2:

Find the solution of the second-order differential equation $16x^2y'' + 8xy' + 3y = 0$ for $0 < x < \infty$.

a) The singular point of the differential equation $16x^2 \cdot y'' + 8x \cdot y' + 3y = 0$ is ___?___

b) The characteristic (or auxiliary) equation is ___?___

Hint: Let $y = x^r$. Then $y' = rx^{r-1}$ and $y'' = (r-1)rx^{r-2}$.

Substitute y, y' and y'' into $16x^2y'' + 8xy' + 3y = 0$: $16x^2\left[(r-1)rx^{r-2}\right] + 8x\left[rx^{r-1}\right] + 3x^r = 0$;

factor out the common term x^r.

c) The roots of characteristic (or auxiliary) equation are ___?___

d) The general solution of the differential equation $16x^2 \cdot y'' + 8x \cdot y' + 3y = 0$ is ___?___

Question 3:

Find the solution of the second-order differential equation $(x+2)^2 y'' - 8(x+2)y' + 8y = 0$ for $-2 < x < \infty$.

a) The singular point of the differential equation $(x+2)^2 \cdot y'' - 8(x+2) \cdot y' + 8y = 0$ is ___?___

b) The characteristic (or auxiliary) equation is ___?___

Hint: Let $y = (x+2)^r$. Then $y' = r(x+2)^{r-1}$ and $y'' = (r-1)r(x+2)^{r-2}$.

Substitute y, y' and y'' into $(x+2)^2 y'' - 8(x+2)y' + 8y = 0$:

$(x+2)^2\left[(r-1)r(x+2)^{r-2}\right] - 8(x+2)\left[r(x+2)^{r-1}\right] + 9(x+2)^r = 0$;

factor out the common term $(x+2)^r$.

c) The roots of characteristic (or auxiliary) equation are ___?___

d) The general solution of the differential equation $(x+2)^2 \cdot y'' - 8(x+2) \cdot y' + 8y = 0$ is ___?___

Question 4:

Find the solution of the second-order differential equation $(x-2)^2 y'' - 5(x-2)y' + 7y = 0$ for $2 < x < \infty$.

a) The singular point of the differential equation $(x-2)^2 \cdot y'' - 5(x-2) \cdot y' + 7y = 0$ is ___?___

b) The characteristic (or auxiliary) equation is ___?___

Hint: Let $y = (x-2)^r$. Then $y' = r(x-2)^{r-1}$ and $y'' = (r-1)r(x-2)^{r-2}$.

Substitute y, y' and y'' into $(x-2)^2 y'' - 5(x-2)y' + 7y = 0$:

$$(x-2)^2\left[(r-1)r(x-2)^{r-2}\right] - 5(x-2)\left[r(x-2)^{r-1}\right] + 7(x-2)^r = 0;$$

Factor out the common term $(x-2)^r$.

c) The roots of characteristic (or auxiliary) equation are ___?___

d) The general solution of the differential equation $(x-2)^2 \cdot y'' - 5(x-2) \cdot y' + 7y = 0$ is ___?___

Question 5:

Find the solution of the second-order differential equation $x^2y'' - 8xy' - 10y = 7x + 2$ for $0 < x < \infty$ with initial conditions $y(1) = 1$ and $y'(1) = 2$.

a) Find the complementary solution (y_c) for the homogeneous equation $x^2y'' - 8xy' - 10y = 0$:

Let $y = x^r$. Then $y' = r \cdot x^{r-1}$ and $y'' = (r-1) \cdot r \cdot x^{r-2}$.

Substitute y, y' and y'' into $x^2y'' - 8xy' - 10y = 0$.

The complementary solution (y_c) is $y_c =$ __?__

b) Finding the particular solution (y_p) for $x^2y'' - 4xy' - 6y = 4x + 5$:

Since the right-hand side is $7x + 2$, we can assume that the particular solution (y_p) has the form $y_p = Ax + B$.

Hence, $(y_p)' = A$ and $(y_p)'' = 0$.

To find y_p, substitute y_p, $(y_p)'$, and $(y_p)''$ into $x^2y'' - 8xy' - 10y = 7x + 2$.

The particular solution (y_p) is $y_p =$ __?__

c) The solution for $x^2y'' - 8xy' - 10y = 7x + 2$ is __?__

Hint: The general solution is $y(x) = y_c(x) + y_p(x)$.

Question 6:

Find the solution of the second-order nonhomogeneous differential equation $x^2y'' - 5xy' + 9y = 2x^2 + 3x + 4, \;\; x > 0.$

a) To find the complementary solution (y_c) for the homogeneous differential equation $x^2y'' - 5xy' + 9y = 0,$ let $y_c = x^r$.

Hence, $y_c' = r \cdot x^{r-1}$ and $y_c'' = (r-1)r \cdot x^{r-2}$.

Substitute y_c, $(y_c)'$, and $(y_c)''$ into $x^2y'' - 10xy' - 12y = 0$: $x^2\left[(r-1)r \cdot x^{r-2}\right] - 5x\left[r \cdot x^{r-1}\right] + 9\left[x^r\right] = 0.$

$y_c =$ ___?___

b) To find the particular solution, y_p, we can use the method of undetermined coefficients and set $y_p = Ax^2 + Bx + C$.

Hence, $(y_p)' = 2Ax + B$ and $(y_p)'' = 2A$.

Substitute y_p, $(y_p)'$, and $(y_p)''$ into $x^2y'' - 5xy' + 9y = 2x^2 + 3x + 4$.

$y_p =$ ___?___

c) The general solution to the differential equation $x^2y'' - 5xy' + 9y = 2x^2 + 3x + 4$ is ___?___

Hint: The general solution is $y(x) = y_c(x) + y_p(x)$.